Business Analytics Summary Report

Dataset: train_data.csv

Generated 2026-08-08 04:42 UTC

200,000	38	5	CatBoost
Dataset Rows	Dataset Columns	Models Compared	Best Model

Executive Summary

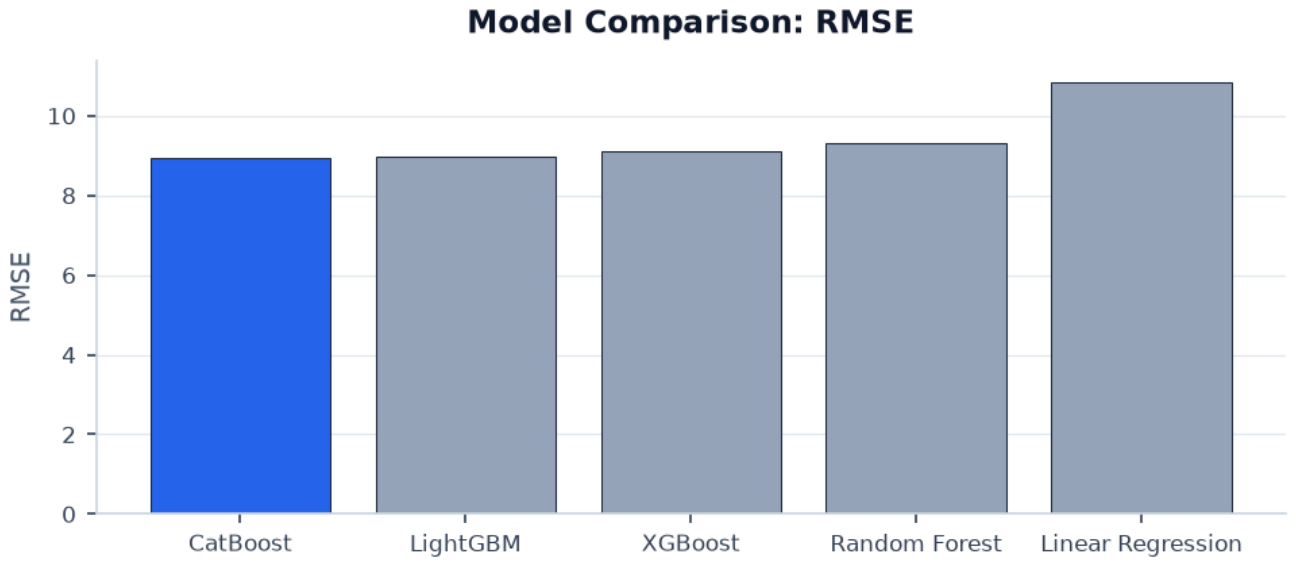
The CatBoost model has been identified as the best performer for predicting risk scores, with a root mean square error (RMSE) of 8.9247 and a coefficient of determination (R2) of 0.9442. This suggests a strong fit between the model's predictions and the actual risk scores. The model's performance is significantly better than the other models compared, including LightGBM, XGBoost, Random Forest, and Linear Regression. The predictions have been generated for 500 rows of data.

Dataset Overview

Metric	Value
Rows	200,000
Columns	38
Missing Cells	315,950 (4.2%)
Duplicate Rows	0
Memory (MB)	91.86

Model Leaderboard

Rank	Model	Training Time (s)	RMSE	MAE	R2	MAPE
1	CatBoost	28.26	8.925	5.745	0.9442	43.27
2	LightGBM	4.347	8.986	5.807	0.9434	44.99
3	XGBoost	2.116	9.107	5.885	0.9419	45.9
4	Random Forest	282.3	9.315	6.009	0.9392	48.13
5	Linear Regression	1.849	10.84	8.17	0.9177	129.1



Best Performing Model: CatBoost

RMSE: 8.925 | MAE: 5.745 | R2: 0.9442 | MAPE: 43.27

Key Insights

The top contributing features to the risk score predictions are related to port congestion levels, geopolitical risk, weather risk, and supplier reliability. The feature importance scores indicate that port congestion level is the most significant factor, with low, medium, and high congestion levels contributing 54.18%, 13.92%, and 12.60% respectively to the model's predictions. Other important features include geopolitical risk index, weather risk index, and supplier reliability score.

Business Recommendations

Based on the model's predictions, businesses can take proactive measures to mitigate potential risks. This includes diversifying suppliers to reduce reliance on a single supplier, investing in weather risk management strategies, and monitoring geopolitical events that may impact operations. By understanding the key factors contributing to risk scores, businesses can develop targeted strategies to minimize their exposure to potential disruptions.

Risk Assessment

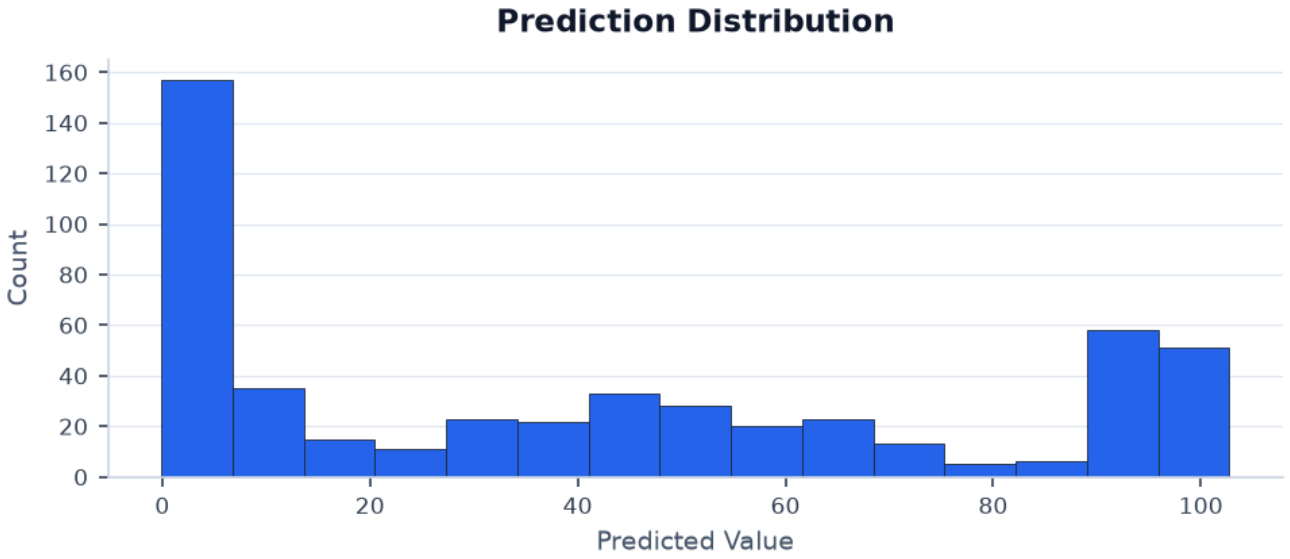
The prediction distribution indicates a wide range of risk scores, with a mean of 41.28 and a standard deviation of 36.19. The minimum and maximum risk scores are -0.08 and 102.74 respectively, suggesting that some predictions have very low risk while others have very high risk. The 25th, 50th, and 75th percentiles of the prediction distribution are 4.56, 35.91, and 71.16 respectively.

Suggested Next Steps

To further refine the model's predictions, it is recommended to explore additional features that may be relevant to risk scores, such as economic indicators or infrastructure development. Additionally, analyzing the predictions for specific segments or groups may provide more targeted insights for business decision-making. The model's performance should also be continuously monitored and updated to ensure that it remains accurate and effective in predicting risk scores.

Predictions Overview

Metric	Value
Total Rows Predicted	500
Average Predicted Value	41.28


Prediction Distribution
Count
160
140
120
100
80
60
40
20
0
0
20
40
60
80
100
Predicted Value